

Energy as Residue - VUPDATED

Corrected finite-frame spectral traversal, aligned to the operator-free TUFT Hopf registry

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Version VUPDATED - July 26, 2026

Status: Corrected canon edition. The original July 18, 2026 paper remains part of the record and is not silently replaced.

Correction notice

This edition corrects Theorem T7 of the original *Energy as Residue* note and narrows three interpretive statements. The central gcd correspondence survives, but the original paper conflated oriented phase matches with energy-level coincidences after the cycle Laplacian identifies a phase with its inverse.

The original statement claimed that the number of ordered eigenmode coincidences was exactly

$$g = \gcd(n_1, n_2).$$

That count is correct for oriented common phases or characters. It is not correct for Laplacian energy coincidences. The corrected count is

$$N_{\text{ordered}} = 2g - \gcd(g, 2),$$

and the number of distinct shared spectral values is

$$N_{\text{distinct}} = \text{floor}(g/2) + 1.$$

The correction was checked by exact arithmetic over all 14,400 ordered frame pairs with

$$1 \leq n_1, n_2 \leq 120.$$

No counterexample occurred.

Abstract

This corrected note retains the three-part traversal of the original paper while separating the operator-free registry from the spectral and physical readings placed upon it. The canonical comparator is the TUFT Hopf direct-limit registry: the compatible family of Hopf bundles

$$S^1 \rightarrow S^{2N+1} \rightarrow CP^N,$$

with standard inclusions between finite stages and direct limit

$$S^\infty \rightarrow CP^\infty.$$

This carrier is a standard, scale-free mathematical object. It supplies phase, fiber action, incidence, winding, projection, and compatibility without selecting a preferred operator or assigning physical units.

A finite cyclic frame is placed inside the fiber group through

$$Z/nZ \simeq \mu_n = z \text{ in } U(1): z^n = 1.$$

The exact registry checksum is

$$\mu_{n_1} \cap \mu_{n_2} = \mu_{\gcd(n_1, n_2)}.$$

Only after this phase-resolved statement is fixed do we introduce the cycle Laplacian. Its spectral reading

$$\lambda(z) = 2 - z - z^{-1}$$

identifies z with z^{-1} . The original T7 error arose precisely at this projection. The corrected theorem separates common phase count, distinct shared spectral values, and ordered eigenmode coincidences. A perturbative splitting result is also retained with the required coupling hypotheses stated explicitly.

The earlier line-current and address-assignment experiments remain recorded, but their interpretation is narrowed. A static counter-wound current with negligible magnetic dipole and large toroidal dipole is called toroidal-dipole-dominant, not a completed anapole demonstration. Uniform address loading is a combinatorial result; it becomes uniform energy density only under an additional equal-energy model. Physical energy enters only after an operator and calibration have been declared.

1. Canonical comparator: the operator-free TUFT registry

1.1 Finite Hopf stages

For each $N \geq 1$, the Hopf bundle is

$$S^1 \rightarrow S^{2N+1} \rightarrow CP^N.$$

The fiber group $U(1)$ acts by common phase multiplication:

$$e^{i\theta}(z_0, \dots, z_N) = (e^{i\theta}z_0, \dots, e^{i\theta}z_N).$$

The standard inclusions

$$(z_0, \dots, z_N) \rightarrow (z_0, \dots, z_N, 0)$$

and

$$[z_0 : \dots : z_N] \rightarrow [z_0 : \dots : z_N : 0]$$

commute with the bundle projections. The finite Hopf stages therefore form a compatible directed family.

1.2 Infinite structured carrier

The direct limit is the universal circle bundle

$$S^1 \rightarrow S^\infty \rightarrow CP^\infty.$$

This is the canonical infinite TUFT comparator used in this edition. It is a structured registry rather than an inert enumeration: its finite stages, inclusions, fiber action, and quotient maps are declared. No Hamiltonian, Beltrami operator, Dirac operator, metric calibration, or physical value is needed to define it.

This terminology also preserves the distinction made elsewhere in the canon between:

- an unstructured infinite registry that merely contains every admissible string; and
- a structured infinite registry whose relations and bonding maps carry mathematical content.

1.3 Finite cyclic frames as fiber subgroups

The finite frame $\mathbb{Z}/n\mathbb{Z}$ is represented inside the phase fiber as the roots-of-unity subgroup

$$\mu_n = \{e^{2\pi i m/n} : 0 \leq m < n\} \subset U(1).$$

The map

$$m \bmod n \rightarrow e^{2\pi i m/n}$$

is a group isomorphism. Thus the finite-frame statements below can be checked first against a standard subgroup of the operator-free Hopf fiber.

2. Direction (a): current on the carrier

The original computation assigned unit current to three line-current configurations and evaluated the magnetic dipole moment m and toroidal dipole moment T :

Configuration	Magnetic dipole magnitude	Toroidal dipole magnitude	Corrected interpretation
Plain ring	12.566	approximately 0	Magnetic-dipole-dominant
(2,5) winding	26.264	5.655	Mixed; magnetic dipole dominates
Counter-wound pair (1,20)+(-1,20)	approximately 2.0×10^{-14}	45.23	Toroidal-dipole-dominant static configuration

The computation refutes the casual claim that a single nested winding is automatically toroidal or anapole-like. Cancellation of the ordinary magnetic dipole requires the counter-transiting construction.

The corrected stopping condition is important. A large toroidal dipole together with a suppressed magnetic dipole does not, by itself, prove a dynamic anapole. A completed anapole claim requires a specified time-dependent current or charge model and demonstration of the relevant radiative-channel cancellation. This paper establishes only the line-current moment hierarchy reported above.

3. Direction (b): assignment and simultaneous closure

Let addresses a in $\mathbb{Z}/n^d\mathbb{Z}$ be assigned to M containers by

$$\sigma(a) = a \bmod M.$$

The loading is uniform exactly when

$$M \text{ divides } n^d.$$

For component periods $p_1, \dots, p_k$, the first simultaneous return is

$$P = \text{lcm}(p_1, \dots, p_k).$$

These are combinatorial closure statements. Equal loading becomes equal energy density only if an additional model assigns equal energy to each address and introduces no position-dependent interaction or weighting. That physical premise is not part of the registry theorem.

4. Direction (c): corrected gcd correspondence

Theorem T7A - Phase registry hears the gcd

Let

$$\mu_n = \{z \in U(1) : z^n = 1\}.$$

For positive integers n_1, n_2 and $g = \gcd(n_1, n_2)$,

$$\mu_{n_1} \cap \mu_{n_2} = \mu_g.$$

Consequently there are exactly g oriented common phases.

Proof. If $z \in \mu_{n_1} \cap \mu_{n_2}$, then the order of z divides both n_1 and n_2 , hence divides g . Thus $z \in \mu_g$. Conversely, every g -th root of unity is both an n_1 -th and an n_2 -th root because g divides n_1, n_2 . Therefore the intersection is exactly μ_g , which has g elements. QED.

This is the operator-free TUFT checksum. It is a statement about the phase registry itself.

Theorem T7B - The Laplacian energy set is the inversion quotient

The cycle-graph Laplacian on $\ell^2(\mathbb{Z}/n\mathbb{Z})$ has eigenvalues

$$\lambda_n(m) = 2 - 2\cos((2\pi m)/(n)).$$

Equivalently, for $z = e^{2\pi i m/n}$,

$$\lambda(z) = 2 - z - z^{-1}.$$

Hence

$$\lambda(z) = \lambda(z^{-1}).$$

For $g = \gcd(n_1, n_2)$, the distinct shared spectral values of the two cycle Laplacians are exactly the distinct spectral values of the cycle of order g :

$$\text{Spec}(C_{n_1}) \cap \text{Spec}(C_{n_2}) = \text{Spec}(C_g)$$

when each side is interpreted as a set rather than a multiplicity-bearing multiset.

The number of distinct shared values is

$$N_{\text{distinct}} = \text{floor}(g/2) + 1.$$

Proof. Equality of spectral values is equivalent to

$$\cos(2\pi x) = \cos(2\pi y),$$

which holds exactly when $x \equiv y \pmod{1}$ or $x \equiv -y \pmod{1}$. The shared phase set before this identification is μ_g by T7A. The Laplacian then quotients that set by inversion $z \sim z^{-1}$. If g is odd, only 1 is fixed and the remaining $g-1$ phases form inverse pairs. If g is even, 1 and -1 are fixed and the remaining $g-2$ phases form inverse pairs. Both cases give $\text{floor}(g/2) + 1$. QED.

Theorem T7C - Correct ordered coincidence count

The number of ordered eigenmode pairs (m_1, m_2) satisfying

$$\lambda_{n_1}(m_1) = \lambda_{n_2}(m_2)$$

is

$$N_{\text{ordered}} = 2g - \gcd(g, 2).$$

Equivalently,

$$N_{\text{ordered}} = \begin{cases} 2g-1, & g \text{ odd}; \\ 2g-2, & g \text{ even}. \end{cases}$$

Proof. The positive congruence

$$(m_1)/(n_1) \equiv (m_2)/(n_2) \pmod{1}$$

has g solutions. The inverse congruence

$$(m_1)/(n_1) \equiv -(m_2)/(n_2) \pmod{1}$$

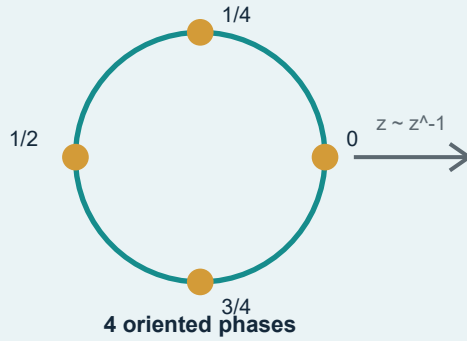
also has g solutions. Their intersection consists of the shared phases fixed by inversion: phase 0 , and also phase $1/2$ when g is even. The intersection therefore has $\gcd(g, 2)$ elements. Inclusion-exclusion gives

$$g + g - \gcd(g, 2) = 2g - \gcd(g, 2).$$

QED.

Worked correction: $n_1=12, n_2=8$

TUFT phase registry -> Laplacian inversion quotient



orbit {0}

spectral value 0 | 1 ordered pair

orbit {1/4, 3/4}

spectral value 2 | 4 ordered pairs

orbit {1/2}

spectral value 4 | 1 ordered pair

Result: 3 distinct spectral values and 6 ordered coincidences, not 4

Here $g=4$. The phase registry has four oriented common phases:

$$0, (1)/(4), (1)/(2), (3)/(4).$$

The Laplacian identifies $1/4$ with $3/4$, leaving three distinct shared spectral values. The ordered coincidences are:

- one pair at phase orbit 0;
- four pairs across the inverse phase orbit $1/4, 3/4$;
- one pair at phase orbit $1/2$.

Thus

$$N_{\text{ordered}}=1+4+1=6,$$

not $g=4$.

5. Perturbative splitting with the required hypotheses

Let the uncoupled operator be

$$H_0 = \begin{bmatrix} L_{n_1} & 0 \\ 0 & L_{n_2} \end{bmatrix},$$

where L_{n_i} is the cycle Laplacian. Let the coupling be purely inter-frame:

$$W = \begin{bmatrix} 0 & V \\ V^* & 0 \end{bmatrix}, \quad H(\varepsilon) = H_0 + \varepsilon W.$$

For a spectral value shared by both blocks, first-order corrections are the eigenvalues of the restriction PWP to the degenerate shared eigenspace. A linear splitting occurs when that restriction is nonzero.

For an isolated, unshared eigenvector supported in one block,

$$\psi, W\psi = 0$$

because W has zero diagonal blocks. Its first-order shift therefore vanishes. Under the usual spectral-gap hypothesis, its leading generic shift is second order.

Accordingly, the original first-order/second-order observation is retained only under these declared conditions:

1. the uncoupled operator is block diagonal;
2. the perturbation is purely inter-frame;
3. the shared eigenspace is coupled nontrivially;
4. the unshared level is isolated from accidental degeneracy;

5. the perturbative spectral gap remains open.

Without these hypotheses, an arbitrary weak perturbation may produce first-order diagonal shifts even for unshared levels.

6. What "energy" means in this edition

Theorems T7A-T7C require no physical energy. T7A is a group-theoretic registry statement. T7B and T7C concern the dimensionless spectrum of a declared combinatorial operator.

To interpret the spectral values as physical energies, an additional operator model and calibration must be supplied. In the terminology of the Hamiltonian calibration supplement:

registry organizes, operator interrogates, calibration measures.

This edition therefore uses "energy as residue" as a Tier II, operator-relative reading. It does not claim that the operator-free registry creates energy, selects a Hamiltonian, or supplies SI scale.

Nor is "residue" used here as an unqualified substitute for every invariant. A formal residue claim in the obstruction calculus must name its representation class and obstruction functional. The exact Tier I content of this paper is the phase-intersection theorem, the inversion quotient, the corrected counting formulas, and the conditional perturbation statement.

7. Recorded fates of the three directions

Direction	Original fate	VUPDATED status
Current moments	Refuted then repaired	Numerical moment hierarchy retained; "anapole" narrowed to toroidal-dipole-dominant static configuration
Address assignment	Formalized	Divisibility and lcm closure retained; energy-density language made conditional
Gcd resonance	Promoted to T7	Core gcd result retained; character, distinct-energy, and multiplicity claims separated and corrected

The correction itself is part of the traversal. A projection from phase to spectral value erased orientation information. The TUFT comparator exposed exactly where that information was lost, and the revised theorem now records both the pre-projection registry object and the post-projection spectral residue.

8. Falsification and reproducibility

This edition fails if any of the following occurs:

1. a pair (n_1, n_2) for which $\mu_{n_1} \cap \mu_{n_2} \neq \mu_{gcd(n_1, n_2)}$;
 2. a pair whose distinct shared Laplacian values differ from the inversion-orbit spectrum of μ_g ;
 3. a pair whose ordered coincidence count differs from $2g \cdot gcd(g, 2)$;
 4. an isolated state under a purely inter-frame perturbation with a nonzero first-order diagonal shift;
 5. a claimed first-order shared splitting when the restricted coupling PWP vanishes.
- The accompanying exact-arithmetic script checks items 1-3 for all 14,400 frame pairs with $1 \leq n_1, n_2 \leq 120$, including the corrected $(12, 8)$ example.

9. Conclusion

The original paper's central intuition survives, but its levels are now separated correctly.

The operator-free TUFT registry retains oriented phase structure. Finite cyclic frames enter that registry as the roots-of-unity subgroups μ_n . Their intersection is governed exactly by the gcd. A cycle Laplacian then reads the phase registry through the inversion-invariant map $z \mapsto z^{-1}$. That reading preserves the shared spectral set but merges inverse phases, changing both the distinct-value count and the multiplicity count.

The corrected chain is therefore

Hopf phase registry $\rightarrow \mu_{n_1} \cap \mu_{n_2} = \mu_g \rightarrow$ operator reading modulo $z \sim z^{-1} \rightarrow$ corrected shared spectrum and multiplicities.

This is the role of the TUFT registry as a canon comparator: later mathematical claims must preserve the distinctions present in the operator-free carrier or state explicitly which distinctions a chosen operator, quotient, projection, or calibration removes.

Correction ledger

Original wording or implication	VUPDATED correction
Ordered energy coincidences equal g	Oriented phase matches equal g ; ordered energy coincidences equal $2g - \gcd(g, 2)$
Shared energy levels counted as g distinct values	Distinct shared values number $\text{floor}(g/2) + 1$
Shared levels always split first order; unshared levels always second order	True only for the declared off-diagonal coupling and nondegeneracy hypotheses
Counter-wound pair is "anapole-class"	It is a toroidal-dipole-dominant static line-current configuration; a dynamic anapole requires additional proof
Uniform address loading means uniform energy density	Only under an additional equal-energy, position-independent model
Energy is residue in the exact formal sense	Retained as a Tier II operator-relative reading; formal obstruction residue requires an explicit R and O

Canon references

- *The Enumerative Resonance Correspondence*, Version 2, July 2026.
- *Energy as Residue*, original traversal, July 18, 2026.
- *Operator-Agnostic Geometry, Topological Obstruction, and Why Completeness Is Not Comprehension*, TUFT Registry Studies Version 2, June 2026.
- *Operator Calibration over the Hopf Registry: Hamiltonian Realization and the Operator Calibration Principle*, Version 1.1, July 2026.
- *The Infinite Registry and Indexed-Residue Proof Engine*, Version 2, July 2026.